

Olist-ic **Recovery**: ROI Through Intelligence

Binary Classification + K- Means Segmentation on the Brazilian E-Commerce Dataset

Maika Diaz | PGD in AI and Machine Learning

Problem Framing

Business problem: ~97% of Olist buyers are one-time purchasers; blanket win-back campaigns waste budget on customers with no repeat intent.

Task type: Supervised binary classification.

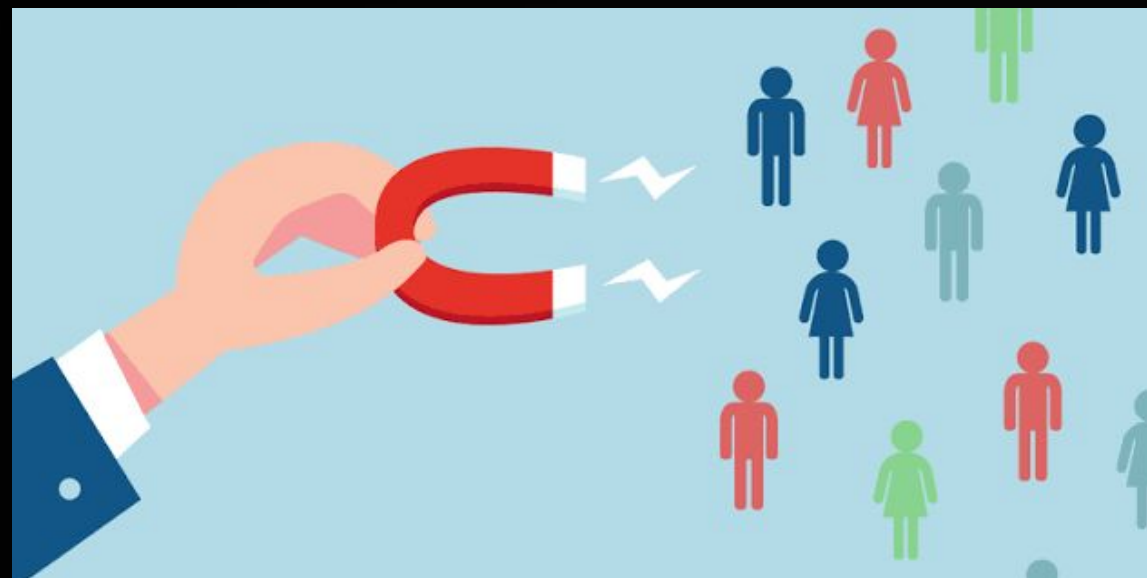
Target variable: target = 1 if customer places a repeat order within 183 days of first order, else 0.

Input features: Signals knowable ~30 days after first-order delivery (spend, cart size, freight, installments, delivery speed/lateness, review score). No post-first-order information used.

Success metrics (technical):

- ROC-AUC — primary metric
- FI-score — secondary, reported due to ~3% class imbalance
- Recall — minimize missed repeat buyers
- Precision/Lift at top-20% threshold — targeting efficiency

Business KPIs: Marketing cost savings vs. blanket campaign, retention uplift, conversion rate per dollar spent.



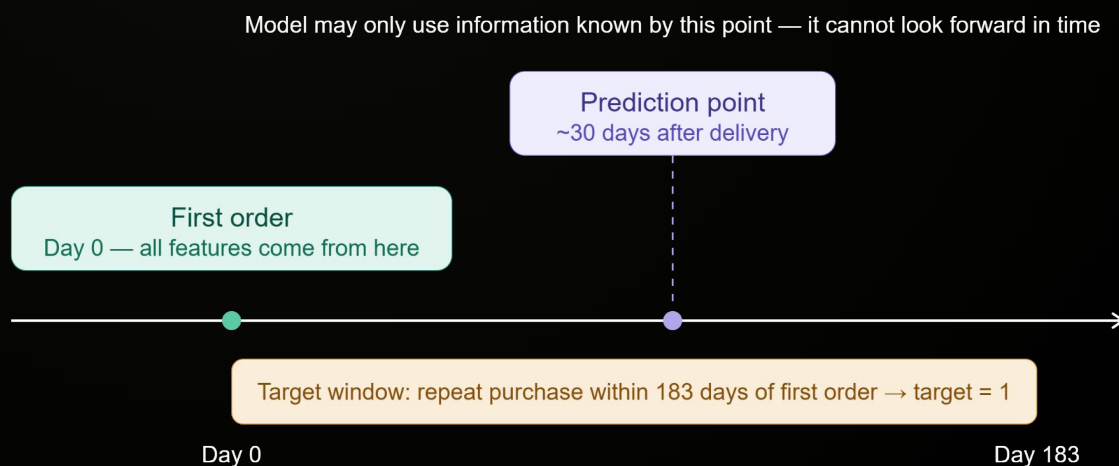
Dataset & Data Dictionary

Feature	Type	Unit	Source
total_spend	float	BRL	order_items
total_freight	float	BRL	order_items
n_items	int	count	order_items
total_paid	float	BRL	order_payments
max_installments	int	count (0-24)	order_payments
review_score	int	1-5 scale	order_reviews
delivery_days	float	days	orders
late	binary	0/1	orders
customer_state	string	ISO state code	customers (audit only, not modeled)
target	binary	0/1	derived

Source: Brazilian E-Commerce
Public Dataset by Olist (Kaggle),
100,000 orders, 2016-2018

- Tables joined: customers, orders, order_items, order_payments, order_reviews
- Missing values: Only review_score had nulls (customers who skipped the survey) → imputed with median (5).
- Outliers: total_spend, total_paid right-skewed (rare high-ticket orders); n_items mostly 1, rare carts up to 20+. Outliers retained — represent legitimate high-value "basket builder" behavior.

Target Construction (Leakage Prevention)



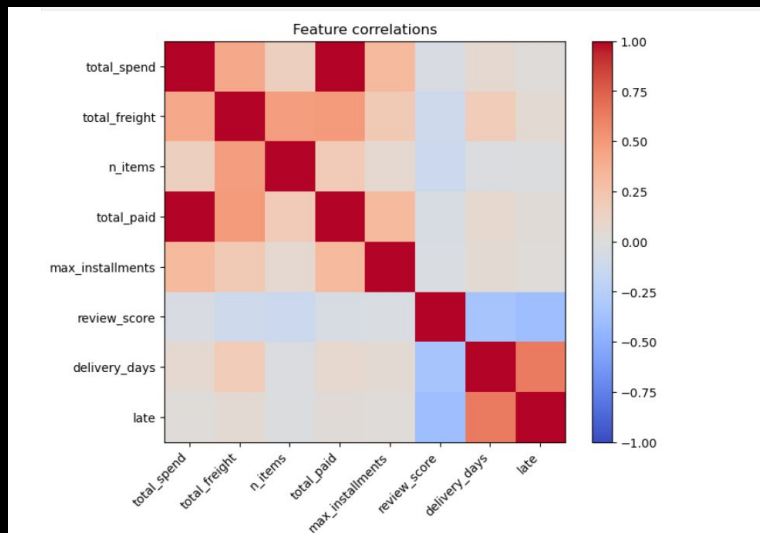
- First purchase timestamp per customer → identify second purchase via order sequence rank.
- $\text{gap} = \text{second_purchase} - \text{first_purchase}$
- $\text{target} = 1$ if $\text{gap} \leq 183$ days else 0
- Right-censoring correction: customers whose first order falls within 183 days of the dataset's max date are excluded - otherwise recent customers get mislabeled as non-repeaters before they've had a fair window to repeat.
- Final dataset: 68,169 customers, 3.1% positive class.

Applied EDA



Distributions:

- $\log(\text{total_spend})$: right-skewed, log-transformed to normalize.
- delivery_days : sharp peak under 20 days, long delayed tail.
- review_score : heavily skewed toward 5-star, secondary spike at 1-star.



Correlations:

- total_spend vs total_paid : $r = 0.996$ (near-perfect redundancy).
- $\text{delivery_days} \leftrightarrow \text{late}$: positively correlated; both negatively correlated with review_score .

Feature Selection & Dimensionality Reduction

- Feature selection (embedded method): Random Forest impurity-based importances used to evaluate feature contribution (see Slide 8).
 - Redundancy identified: total_spend and total_paid are collinear ($r=0.996$) - total_paid is a candidate for removal without loss of predictive power.
 - Dimensionality reduction (PCA): 4–5 principal components required to capture 80–90% of variance. Final models retain all 8 original features - dimensionality was low enough that interpretability was prioritized over compression.
-

Model Implementation & Comparison

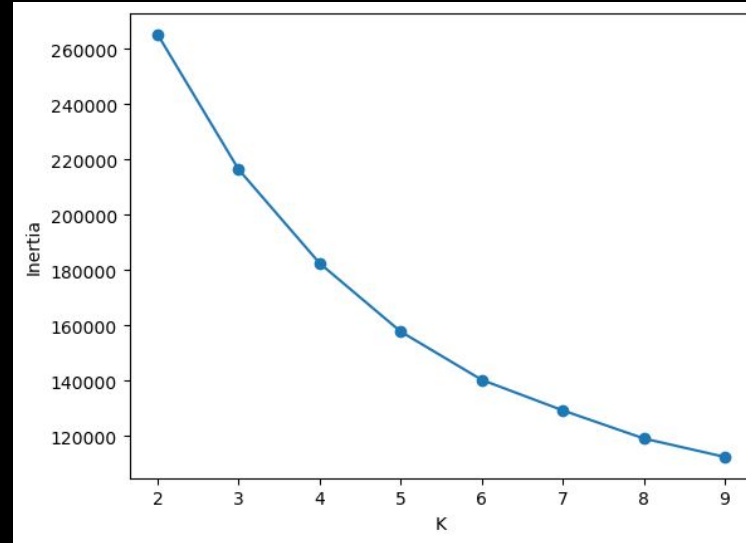
Model	AUC	F1 (default threshold)
Logistic Regression (balanced)	0.5477	0.0667
Random Forest — default	0.5344	0.0
Random Forest — constrained	0.5597	0.0757

Stratified 80/20 train-test split, random_state=42.

Finding: Default Random Forest overfits and collapses to predicting zero positives on the test set (F1=0.0). Constraining tree depth and leaf size produces the best-generalizing model.

Selected model: Constrained Random Forest.

K-Means: Elbow & Selection



Task type: Unsupervised clustering - separate from the classification task, run on a distinct feature subset (total_spend, n_items, max_installments, review_score, delivery_days), standardized before clustering.

Model selection (elbow method): Inertia plotted for K=2 through K=9. Elbow observed at K=4 - diminishing inertia reduction beyond this point.

Selected: K=4.

K-Means: Segment Profiles

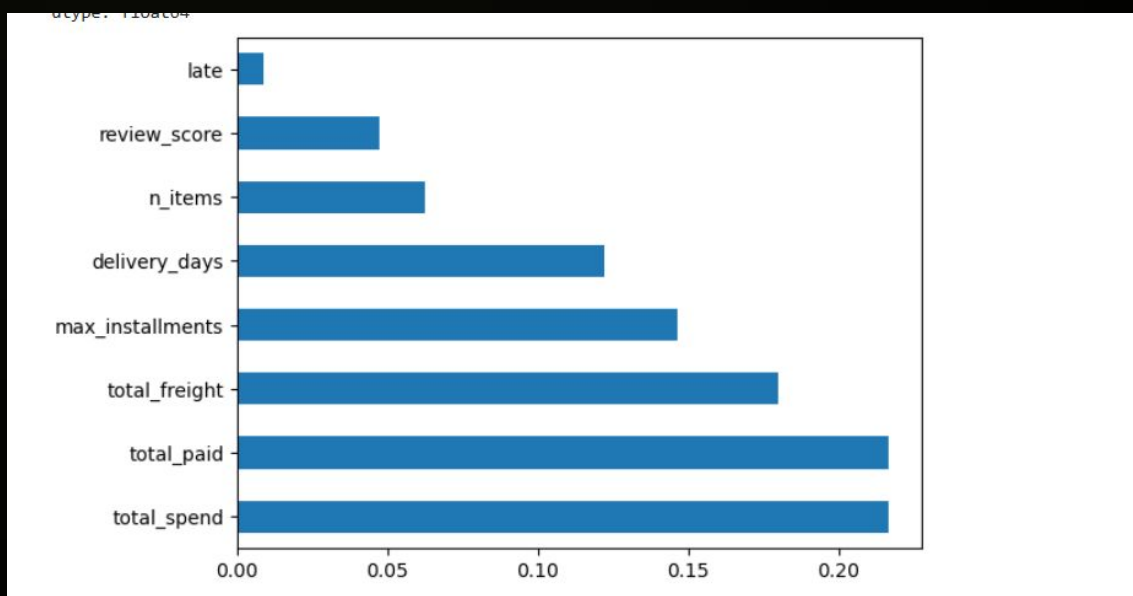
K-Means segment profiles (K=4) — cluster means on unscaled first-order features

Segment (cluster id)	n	Mean spend (R\$)	Mean items	Mean installments	Mean review	Mean delivery (days)	Repeat rate
Casuals (0)	44,155	92.10	1.06	1.87	4.62	10.97	2.87%
Big Spenders (1)	10,221	321.21	1.11	7.89	4.39	12.72	3.59%
Burned by Delivery (2)	9,994	118.75	1.13	2.61	1.64	24.59	2.87%
Basket Builders (3)	1,473	296.85	3.83	3.78	3.37	12.47	5.70%

Key finding: Basket Builders (n=1,473, the smallest segment) show the highest repeat rate at 5.70% - roughly double the marketplace baseline (~2.87–3.59% across the other three segments) - despite not being the highest spenders. This is the segmentation-side confirmation of the `n_items` explainability gap flagged in the feature importance slide: cart size predicts repeat behavior more strongly than transaction value does.

Explainability

Method: Impurity-based (Gini) feature importances - embedded feature selection method.



Findings:

- total_spend and total_paid split importance almost evenly (~0.217 each) due to their 0.996 correlation - model divides credit between two measurements of the same underlying value.
- n_items shows low global importance despite basket size being strongly predictive within the "Basket Builder" segment (2× baseline repeat rate) - a global-importance blind spot; feature is rare but locally powerful

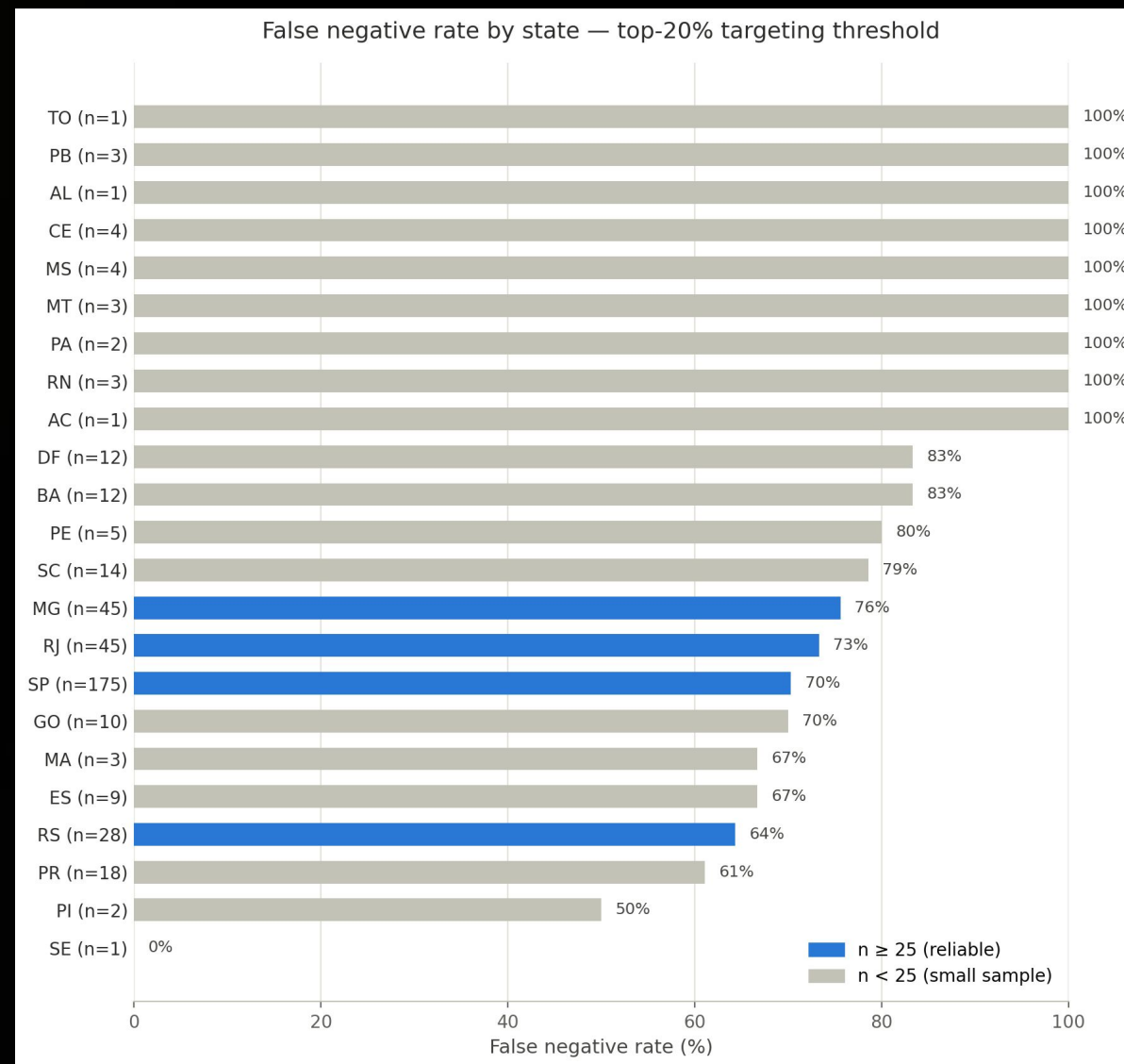
Bias & Fairness Audit

Method: False Negative Rate (FNR) by customer_state, computed on actual repeaters in the test set. customer_state excluded from model training - used only for audit.

Threshold sensitivity finding: At threshold=0.05, FNR $\approx$ 0% everywhere - an artifact of flagging nearly all customers as repeaters, not genuine fairness. Re-audited at the realistic top-20% operating threshold.

Results at top-20% threshold:

- Large states with reliable sample sizes: São Paulo (n=175), Rio de Janeiro & Minas Gerais (n=45) - FNR clustering 70–76%.
- Small states (e.g., Sergipe, n=1): FNR results statistically unreliable due to sample size



Model Limitations

- Class imbalance: ~3.1% positive rate - addressed via `class_weight=balanced` and F1/AUC over accuracy, but not via resampling (SMOTE, undersampling) - untested alternative.
- Leakage control: Enforced via first-order-only features and the 30-day prediction cutoff; right-censoring correction applied to target construction.
- Overfitting: Demonstrated directly by default Random Forest (F1=0.0); mitigated via depth/leaf constraints, not cross-validated hyperparameter search.
- Explainability gap: No SHAP/LIME/PDP/ICE - impurity importances only
- Model performance ceiling: AUC 0.53-0.56 across all three models; indicates the available first-order features have limited separative power; repeat purchase behavior is only weakly predictable from a single early transaction.

Recommendations (next steps):

- Compute SHAP values on the constrained Random Forest to resolve the total_spend/total_paid collinearity split and the n_items global-vs-local importance gap.
- Add demographic parity and disparate impact ratio to the fairness audit; consider per-region threshold calibration if disparity holds under a larger sample
- Test SMOTE/undersampling against the current class_weight approach
- Cross-validate hyperparameters (max_depth, min_samples_leaf) rather than the current single manual configuration
- Silhouette score for the K=4 clustering choice, as a second validation alongside the elbow plot